What are IP, MAC, DNS, DHCP and ports? Include a table of 12 common ports: 21, 22, 23, 25, 53, 80, 110, 143, 443, 445, 3306, 3389 with service and typical risk.

Computer networks allow devices to communicate and share information. To understand how this communication works, it is important to learn about IP addresses, MAC addresses, DNS, DHCP, and ports.

These concepts are essential for network troubleshooting, cybersecurity monitoring, and identifying potentially exposed services.

What Is an IP Address?

An **IP (Internet Protocol) address** is a logical address assigned to a network interface. It helps identify the interface and allows data packets to be routed between networks.

There are two main versions:

- **IPv4**: Uses 32 bits, commonly written as four decimal numbers separated by dots. **Example**: 192.168.1.10.
- **IPv6**: Uses 128 bits and is written in hexadecimal groups. **Example**: 2001:db8::10.

IP addresses can be private or public.

- **Private IP**: Used within private networks, such as a home network. Example: 192.168.1.10.
- **Public IP**: Used for communication across the public internet and assigned according to internet addressing arrangements.

Example: A laptop may have a private IP address that allows it to communicate with other devices on the same home network.

What Is a MAC Address?

A **MAC (Media Access Control) address** is a link-layer identifier associated with a network interface. Ethernet and Wi-Fi networks use MAC addresses to deliver frames across a local network.

Example: 00:1A:2B:3C:4D:5E

Important points:

- MAC addresses are commonly represented in hexadecimal.
- They are used for local network communication.
- Network devices use IP addresses for logical addressing and routing, while MAC addresses help deliver frames on the local link.
- A MAC address can sometimes be changed or randomized, so it should not be treated as proof of a device's identity.

Example: When two computers communicate on the same Ethernet network, their network interfaces use MAC addresses to deliver local frames.

What Is DNS?

DNS (Domain Name System) translates domain names into information that computers can use, including IP addresses.

For example, when a user enters example.com into a browser, DNS can help find the IP address associated with that domain.

Without DNS, users would often need to remember numerical IP addresses instead of convenient domain names.

Common DNS record types

Record	**Purpose**
A	Maps a name to an IPv4 address.
AAAA	Maps a name to an IPv6 address.
MX	Identifies mail servers for a domain.
CNAME	Provides an alias for another domain name.
TXT	Stores text information, often used for verification and email security policies.

NS	Identifies authoritative name servers for a domain.

Security risk: A compromised or misconfigured DNS environment may redirect users, expose internal information, or support other attacks.

What Is DHCP?

DHCP (Dynamic Host Configuration Protocol) automatically provides network configuration to devices.

It can supply:

- An IP address.
- A subnet mask or network prefix.
- A default gateway.
- DNS server addresses.
- Lease information indicating how long the assignment is valid.

How DHCP works

The common DHCPv4 process is called DORA:

1. Discover: The client searches for a DHCP server.
2. Offer: A server offers network configuration.
3. Request: The client requests the offered configuration.
4. Acknowledge: The server confirms the assignment.

Example: When a laptop connects to home Wi-Fi, the router's DHCP service commonly assigns it an IP address automatically.

Security risk: A rogue DHCP server may provide incorrect network settings and redirect traffic. Network protections can help prevent unauthorized DHCP services.

What Is a Port?

A port is a transport-layer number used by TCP or UDP to identify a communication endpoint associated with an application or service.

Port numbers range from 0 to 65535.

Examples:

- TCP port 22 is commonly used by SSH.
- TCP port 443 is commonly used by HTTPS.
- TCP port 3306 is commonly used by MySQL.

A port number alone does not guarantee that a particular service is running. Administrators should verify actual listening services and their configurations.

12 Twelve Common Ports and Their Security Risks

Port	Common service	Typical security risk
21	FTP	Traditional FTP may transmit credentials and data without encryption.
22	SSH	Weak passwords, stolen keys, or unpatched SSH software may allow unauthorized access.
23	Telnet	Credentials and data may be transmitted in plaintext.
25	SMTP	An incorrectly configured mail server may permit unauthorized relaying or be abused to send spam.
53	DNS	Misconfigured DNS servers may enable information leakage or DNS amplification abuse.
80	HTTP	Traffic is generally unencrypted, and vulnerable web applications may be exposed.
110	POP3	Email credentials may be exposed if encryption is not used.
143	IMAP	Email access may be vulnerable to credential interception without TLS protection.
443	HTTPS	Encryption protects data in transit, but application vulnerabilities and weak TLS settings remain possible.
445	SMB	Exposed file sharing, weak permissions, or unpatched SMB services can

		enable serious attacks.
3306	MySQL	Publicly exposed databases may be targeted through weak credentials or software vulnerabilities.
3389	RDP	Weak credentials or vulnerable remote desktop services may enable unauthorized remote access.

Important security practices

- Disable unnecessary services and close unused ports.
- Restrict access to services using firewalls.
- Use encrypted protocols such as SSH instead of Telnet.
- Configure secure versions of email and file-transfer protocols.
- Keep services patched and use strong authentication.
- Avoid exposing databases and remote administration services directly to the public internet unless there is a justified, securely configured requirement.

Difference Between IP, MAC, DNS, DHCP and Ports

Concept	Main purpose	Example
IP address	Logical addressing and routing.	192.168.1.10
MAC address	Local link-layer frame delivery.	00:1A:2B:3C:4D:5E
DNS	Resolves names into records such as IP addresses.	Resolving example.com
DHCP	Automatically supplies network settings.	A router assigns an IP to a laptop.
Port	Identifies a transport-layer communication endpoint.	TCP port 443 for HTTPS

Useful Linux Commands for Networking

Command	Purpose
ip a	Displays network interfaces and IP addresses.

ip route	Displays routes and the default gateway.
ping example.com	Tests connectivity using ICMP where permitted.
nslookup example.com	Queries DNS.
dig example.com	Retrieves DNS information.
ss -tulpn	Displays listening TCP/UDP sockets and, where permissions allow, associated processes.
traceroute example.com	Helps investigate the network path to a destination.

Some commands may need to be installed separately, and network firewalls may block certain tests.

Conclusion

IP addresses support logical addressing and routing, MAC addresses support local network delivery, DNS resolves domain names, DHCP automates network configuration, and ports direct transport traffic to application endpoints.

Understanding these concepts helps cybersecurity professionals troubleshoot network problems, inspect exposed services, investigate suspicious traffic, and reduce unnecessary network exposure.